MARTIN KRUTSKÝ

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Passionate about responsible and ethical research of AI through neurosymbolic and XAI techniques.

RESEARCH INTERESTS

- **Mechanistic Interpretbility:** Neurosymbolic AI for prescriptive mechanistic interpretability.
- **Explainability for Responsible AI:** Assessment of explainability techniques for AI regulation.

RESEARCH EXPERIENCE

IDA Intelligence Data Analysis Lab, CTU • **AI Researcher & PhD Student** since 2024
Integrating neurosymbolic methods into circuit discovery and sparse feature learning.

Center for Machine Perception, CTU • **Student Researcher** 2018–2019
Implemented CNN/OpenFace-based recognition of nationality and first impressions.

INDUSTRY EXPERIENCE

Leeaf/McEDISON • **ML Researcher & AI Engineer** since 2023
Designing tabular ML models (IVF treatment, DB data) & LLM apps (RAG, tools, etc.).

Microsoft • **Software Engineering Intern** 2022
Modeled MS Teams' middleware server load for improved Azure resource planning.

Blindspot Solutions • **Machine Learning Developer** 2020–2022
Developed NER & classification modules (Transformers), chatbots, and pricing models.

Barclays • **Part-Time Developer** 2019–2020
Developed activity prediction system (scikit-learn) and monitoring dashboard.

EDUCATION

- Czech Technical University • **Ph.D. in Artificial Intelligence** since 2024
  Research topic: *Explainable Neurosymbolic AI* • Advisor: Gustav Šír
- Czech Technical University • **M.Sc. in Artificial Intelligence** 2021–2024
  Master Thesis: *Interpretable Symmetry-Aware Deep Learning for Planning*
  Graduated with honors; Grade: **1.26 ECTS ≈ 3.74 GPA**
- Aalto University • Erasmus+ exchange studies 2023
- prg.ai Minor • Selective interdisciplinary curriculum 2018–2021

SERVICE AND MEMBERSHIP

- Czech Alpha Chapter of Upsilon Pi Epsilon • **President** since 2024
  Representation, event organization, and communication on behalf of the chapter
- Responsible AI at Czech Technical University • **Founding Member** since 2023
  Science communication, interdisciplinary research, and event coordination

SCHOLARSHIPS

- Upsilon Pi Epsilon • **UPE Academic Achievement Award** 2022
- Czech Technical University • **Scholarship for Talented Students** 2018

SELECTED PUBLICATIONS (GOOGLE SCHOLAR)

- **Geometric Deep Learning for the Rubik's Cube Group** (Krutský, Šír) 2025
  IEEE Transactions on Neural Networks and Learning Systems (Q1)
- **Binarizing PI-GNNs for Combinatorial Optimization** (Krutský et al.) 2025
  28th European Conference on Artificial Intelligence (ECAI)
- **XAI Desiderata for Trustworthy AI: Insights from the AI Act** (Krutský et al.) 2025
  The European Workshop on Trustworthy AI (at ECAI 2025)

SELECTED PRESENATIONS AND TALKS

- When AI Slanders Us & Evaluating Explanations (Němeček, Krutský, Peleška) 2025
  Public talks at AI Days 2025
- Assessing XAI Methods for AI Safety Governance (Krutský, Němeček, Gürtler, Peleška) 2025
  Oral presentation at the International Conference on Large-Scale AI Risks
- Dimensions of Explainability in AI Alignment (Krutský, Němeček, Peleška, Gürtler) 2025
  Poster presentation at the 29th International Student Scientific Conference (POSTER)
- Ethics in AI Development (Němeček, Krutský) 2024
  Public talk for the AI 4 All event at AI Days 2024
- Responsible AI in Prague (Krutský) 2023
  Public talk at the AI Days 2023

DATA STACK

ML Frameworks:	PyTorch | PyTorch Geometric | Transformers | Scikit-Learn | XGBoost
Generative AI:	LLMs | RAG | tools & agents | guardrails | Ollama | speech-to-text models
Data Science:	Pandas | NumPy | Seaborn | Matplotlib | basics of Hadoop, Spark

SW STACK

MLOps/DevOps:	MLFlow & Ray Tune | Docker | Digital Ocean | AWS | Git + CI/CD | JIRA
Web Stack:	FastAPI & Pydantic | Flask & Marshmallow | basics of Django, Next.js
Programming:	Python | Java | SQL | basics of JavaScript, C++, Kotlin

SUMMER SCHOOLS

Eastern European ML Summer School	2025
Human-aligned AI Summer School	2024
AI & Ethics Summer School	2022

LANGUAGES

English	●●●●◐	Proficient
German	●●◐○○	Intermediate
Czech	●●●●●	Native